Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Backend Software Engineer** position with **Fanatics Live**. I bring over three years of professional experience building backend systems, distributed services, and production-grade APIs, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production backend systems that directly reflect the technical depth this role demands — scalable **RESTful APIs**, **event-driven architectures**, **asynchronous** processing pipelines, and **real-time** systems handling high-concurrency traffic. As a Software Engineer at Sperse, I designed **event-driven**, **real-time** backend services in **Python** and **FastAPI** routing requests across 40+ specialized sub-agents with tenant-aware **JWT** access control and **OpenTelemetry** instrumentation, serving 10,000+ concurrent users. Alongside it, I built an **asynchronous** workflow orchestration platform using **Temporal** and **Redis** for durable, resumable execution of **distributed** long-running tasks — work that maps directly to the order processing, inventory orchestration, and livestream interaction workflows at the core of Fanatics Live's platform. I also built a production **FastMCP** server exposing 160+ **RESTful API** endpoints with **Pydantic** schema validation, **OpenTelemetry** monitoring and logging, and **secure** multi-tenant access control. To keep these systems production-ready, I built GitHub Actions **CI/CD** pipelines with trace-based eval suites that flag latency regressions and unsafe behavior before every deployment. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to backbone infrastructure that users relied on daily. I designed a cloud-agnostic **Storage Service** in **Java** and **Spring Boot** with **PostgreSQL** and **Redis** handling file operations across **AWS S3**, **Google Cloud Storage**, and **MinIO** via expiring signed URLs — hands-on experience with cloud-native, multi-environment backend design. I also orchestrated **event-driven** long-running jobs with **AWS Step Functions**, **Lambda**, and **message queues**, improving error recovery and **observability** across **100K+** weekly executions, and optimized **PostgreSQL schema design** with window functions, materialized views, and composite indexes while contributing to **incident response** workflows. Together, these roles have prepared me to independently deliver production-quality backend services, contribute meaningfully to system design and platform reliability, and support operational excellence in a fast-moving environment.

My academic research and side-project work have given me a rigorous foundation in **distributed systems** design and performance-driven engineering. At the Pace Artificial Intelligence Lab, I built systems that demanded careful experimental design, deep performance analysis, and the ability to reason about distributed execution patterns at scale. I also built Dark Factory — a **distributed** coding-agent orchestrator in **Elixir** and **TypeScript** that decomposes large requests into dependency-aware task **DAGs**, executes them across multi-machine agents with **asynchronous** retries, and propagates **real-time** status across distributed nodes. This work sharpened my ability to design fault-tolerant, high-throughput **distributed systems** — precisely the skills needed to build and scale the livestream commerce and inventory orchestration systems at Fanatics Live.

I have always been passionate about fan obsession and building experiences that amplify pride and create connections for all fans, and I envision myself contributing my talent to Fanatics Live's mission of making every fan interaction — from a live drop to a real-time purchase — feel fast, reliable, and community-driven. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan